Confidence Calibration under Ambiguous Ground Truth

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Abstract

Confidence calibration implicitly assumes a unique ground-truth label per input. When multiple annotators genuinely disagree—as in medical diagnosis, natural language inference, and perceptual tasks—this assumption breaks down. We show that post-hoc calibrators fitted on majority-voted labels exhibit a systematic *calibration gap*: they appear well-calibrated against voted labels yet remain substantially miscalibrated against labels drawn from the annotator distribution. The gap is structural, affecting Temperature Scaling, Platt scaling, and Histogram Binning alike, because all share the same one-hot target. We prove that Temperature Scaling is biased toward lower temperatures than the annotator distribution requires, and that the gap grows monotonically with annotation entropy. To close it, we propose **Soft-Label Temperature Scaling (SLTS)** and related methods that optimize proper scoring rules against the full annotator distribution. Experiments on CIFAR-10H, ChaosNLI, ISIC 2019, and DermaMNIST (the last two using synthetic annotator models calibrated to published clinical agreement rates)—spanning seven architectures across vision and language—show that our methods reduce true-label ECE by 48–91% relative to standard Temperature Scaling.

1 Introduction

A well-calibrated classifier $f : \mathcal{X} \to \Delta^K$ produces probability vectors $\hat{p}(x) = f(x)$ such that stated confidence matches empirical accuracy: when $\hat{p}_{\hat{c}}(x) = p$ for the predicted class $\hat{c} = \arg\max_k \hat{p}_k(x)$, the true probability of $\hat{c}$ being correct is exactly p. Formally,

$$\mathbb{P}\big(\hat{Y} = Y \mid \hat{p}_{\hat{Y}}(X) = p\big) = p, \quad \forall\, p \in [0, 1], \tag{1}$$

where Δ^K denotes the K-dimensional probability simplex. Calibration is critical for safe deployment in medicine [Kompa et al., 2021], autonomous driving, and other high-stakes settings.

The hidden assumption. Equation (1) implicitly assumes a unique ground-truth label Y for every input x. In practice this assumption frequently fails: a skin lesion may be legitimately classified as malignant or benign by different dermatologists; a natural-language premise may or may not entail a hypothesis depending on pragmatic context; low-resolution objects can be categorically ambiguous. In such settings the correct target is not a single label but a *distribution over labels*, $\pi(\cdot \mid x) \in \Delta^K$, reflecting irreducible aleatoric uncertainty shared by rational annotators.

The standard practice and its failure. Standard practice aggregates annotations by majority vote to obtain a *voted label* y^* and calibrates against it. We show this produces a systematic *calibration gap*: the calibrator achieves low ECE against voted labels ($\mathrm{ECE}_{\text{voted}}$) while remaining substantially miscalibrated against true labels drawn from the annotator distribution ($\mathrm{ECE}_{\text{true}}$, following Stutz et al.

[2023]). The mechanism is straightforward: ambiguous examples appear underconfident relative to the one-hot voted target, so Temperature Scaling selects a lower T than the annotator distribution requires, amplifying overconfidence. This failure is universal—Temperature Scaling, Platt scaling, and Histogram Binning all *widen* the gap in a controlled experiment (Section 4)—because the root cause is the voted-label target, not method capacity.

Contributions.

1. **Formal framework** (Section 3): we define true-label calibration and the calibration gap Δ with respect to the annotator distribution $\pi(\cdot \mid x)$.
2. **Motivating example** (Section 4): a controlled 3-class experiment in which all standard calibrators *widen* the gap, from $+6.6$ to $+7.3$–$+8.0$ pp.
3. **Theory** (Section 5): we prove that TS is biased toward lower temperatures than the annotator distribution requires, and that the gap grows monotonically with annotation entropy.
4. **Methods** (Section 6): post-hoc calibrators—SLTS, MCTS, VS, IR-Soft, SoftPlatt, and Dirichlet-Soft—that optimize proper scoring rules against annotator distributions; we further show (Section 7.4) that the improvement is due to the soft target, not additional model capacity.
5. **Experiments** (Section 7): evaluation on four benchmarks (CIFAR-10H, ChaosNLI, ISIC 2019, DermaMNIST in appendix) across seven architectures spanning vision and language, confirming the gap is modality-agnostic.

2 Related Work

Confidence calibration. Guo et al. [2017] showed that modern networks are miscalibrated and that TS is an effective post-hoc remedy. Isotonic regression [Zadrozny and Elkan, 2002], Platt scaling [Platt, 1999], and Dirichlet calibration [Kull et al., 2019] extend this to multiclass settings—all assuming a unique ground-truth label. Minderer et al. [2021] showed ViTs are better calibrated than CNNs; Bai et al. [2021] and Tian and Xiao [2023] revisit the sufficiency of TS. All share the voted-label assumption; we show this assumption—not method capacity—is the limiting factor. Label smoothing [Szegedy et al., 2016, Müller et al., 2019] applies uniform soft targets during training; we instead use annotation-informed soft labels for post-hoc calibration requiring no retraining.

Ambiguous ground truth. Dense annotation datasets [Peterson et al., 2019, Nie et al., 2020] and multi-annotator learning [Uma et al., 2021, Rodrigues and Pereira, 2018] have received growing attention. Baan et al. [2022] show ECE is theoretically ill-defined when humans genuinely disagree and propose instance-level alternatives; our work complements this by providing methods that close the resulting calibration gap. Gordon et al. [2022] and Plank [2022] highlight epistemological problems with annotation aggregation in NLP. Giulimondi et al. [2024] use disagreement as an abstention signal; Baan et al. [2025] show training on soft labels preserves epistemic uncertainty—a training-time complement to our post-hoc approach.

Conformal prediction under ambiguous ground truth. Stutz et al. [2023] show that conformal sets calibrated on voted labels undercover the true annotator distribution—the analogous failure for set prediction that our work addresses for point prediction. Calibration under covariate shift [Park et al., 2020, Wang et al., 2020] is orthogonal to our label-ambiguity setting.

3 Problem Formulation

3.1 Setup and Notation

Let $\mathcal{Y} = \{1, \ldots, K\}$ be the label space. A classifier $f : \mathcal{X} \to \Delta^K$ outputs $\hat{p}(x) = \mathrm{softmax}(z(x))$ from pre-softmax logits $z(x) \in \mathbb{R}^K$. The **annotator distribution** $\pi(\cdot \mid x) \in \Delta^K$ is estimated from m annotations as $\hat{\pi}_k(x) = \frac{1}{m}\sum_j \mathbf{1}[a_j = k]$. The **voted label** is $y^* = \arg\max_k \hat{\pi}_k(x)$. Standard calibration uses voted labels; Temperature Scaling [Guo et al., 2017] minimizes

$$\mathcal{L}_{\text{TS}}(T) = -\frac{1}{n}\sum_{i=1}^{n} \log \mathrm{softmax}(z_i/T)_{y_i^*}. \tag{2}$$

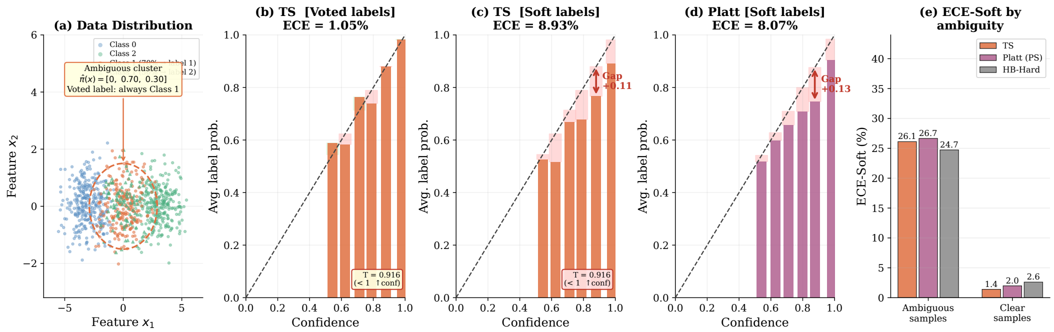

Figure 1: **Motivating example: all standard calibration methods fail.** **(a)** Data distribution: the middle cluster is ambiguous ($\hat{\pi} = [0, 0.70, 0.30]$; voted label always Class 1). **(b)** TS against voted labels: appears well-calibrated ($\text{ECE}_{\text{voted}} = 1.1\%$). **(c)** TS against true labels: large calibration gap ($\text{ECE}_{\text{true}} = 9.1\%$, $\Delta = +8.0$ pp). **(d)** Platt scaling against true labels: also fails ($\text{ECE}_{\text{true}} = 8.3\%$), confirming the gap is independent of method capacity. **(e)** Stratified ECE_{true}: the calibration gap is concentrated in the ambiguous cluster for all voted-label methods; see Table 1 for the full ECE-Voted vs. ECE-True breakdown.

3.2 True-Label Calibration and the Calibration Gap

Following Stutz et al. [2023], we distinguish the voted label y^* from the **true label** $\tilde{y} \sim \pi(\cdot \mid x)$, drawn from the annotator distribution.

Definition 1 (True-Label Calibration)**.** *f is **true-label calibrated** if for all $p \in [0, 1]$: $\mathbb{P}(\tilde{Y} = \hat{c}(X) \mid \hat{p}_{\hat{c}}(X) = p) = p$, where $\hat{c}(x) = \arg\max_k \hat{p}_k(x)$ and $\tilde{Y} \sim \pi(\cdot \mid X)$.*

The **voted-label ECE** $\text{ECE}_{\text{voted}}$ is the standard ECE against y^*. The **true-label ECE** ECE_{true} averages ECE over 100 draws $\tilde{y}_i \sim \pi(\cdot \mid x_i)$ ($B = 15$ equal-width bins). The **calibration gap** $\Delta = \text{ECE}_{\text{true}} - \text{ECE}_{\text{voted}}$ is positive when the model appears well-calibrated on voted labels but remains overconfident against the annotator distribution.

4 Motivating Example

Setup. A 3-class 2D Gaussian dataset: class 0 is always labeled 0 ($\pi_0 = 1$); class 1 has annotator distribution $\pi(x) = [0, 0.70, 0.30]$ (voted label always 1); class 2 is always labeled 2 ($\pi_2 = 1$). A 2-layer MLP (128 hidden units) is trained on voted labels. This is the simplest setting that exposes the calibration gap: one cluster is perfectly unambiguous to the annotators but the voted label is always the majority class.

Results. Table 1 and Figure 1 show that the calibration gap is a structural consequence of the voted-label target. TS achieves $\text{ECE}_{\text{voted}} = 1.1\%$ but $\text{ECE}_{\text{true}} = 9.1\%$ ($\Delta = +8.0$ pp)—*larger* than the uncalibrated gap of $+6.6$ pp. TS sets $T = 0.92 < 1$, boosting confidence toward the voted target and thereby pushing the model *further* from the true annotator distribution. Platt scaling ($\Delta \approx +7.3$ pp) and even non-parametric Histogram Binning ($\Delta = +6.6$ pp) exhibit the same pathology, confirming that the failure lies in the target, not the method.

5 Theory: Why Standard Calibration Fails

We now formalize two complementary aspects of the calibration gap: the *direction* of the TS bias (Proposition 1) and its *scaling* with label ambiguity (Proposition 2).

Proposition 1 (Direction of TS Bias)**.** *Consider a calibration set containing an ambiguous cluster in which the voted label y^* equals the majority class for every example, but the true annotator probability of y^* is $\pi_{y^*}(x) = q < 1$. If the model is already reasonably accurate on this cluster ($\text{softmax}(z_i)_{y^*} > q$), then TS minimizing $\mathcal{L}_{TS}$ finds $T^* < 1$ (increasing model confidence), whereas soft calibration requires $T^* > 1$ (reducing confidence to q).*

Table 1: **Motivating toy example: all voted-label calibration methods widen the calibration gap.** Every voted-label method reduces $\mathrm{ECE}_{\text{voted}}$ but *increases* $\mathrm{ECE}_{\text{true}}$, growing the gap $\Delta = \mathrm{ECE}_{\text{true}} - \mathrm{ECE}_{\text{voted}}$ from $+6.6$ pp (uncalibrated) to $+7.3$–$+8.0$ pp. TS sets $T = 0.92 < 1$ (wrong direction for ambiguous data). The failure is structural: it stems from the voted-label target, not method capacity.

Method	T	$\mathrm{ECE}_{\text{voted}}$ (%) ↓	$\mathrm{ECE}_{\text{true}}$ (%) ↓	Gap Δ (%)
Uncalibrated	—	1.6	8.2	+6.6
TS	0.92	**1.1**	9.1	+8.0
Platt (PS)	—	1.0	8.3	+7.3
HB-Hard	—	1.6	8.2	+6.6

Proof sketch. On the ambiguous cluster, all voted labels are y^*, so the voted-label loss (2) is minimized by pushing $\mathrm{softmax}(z_i/T)_{y^*} \to 1$, i.e. $T \to 0$. Hence $\partial\mathcal{L}_{\text{TS}}/\partial T > 0$ and the optimal $T^* < 1$. Soft calibration instead minimizes $\mathcal{L}_{\text{SLTS}}$ with target $q < 1$; the minimum requires $\mathrm{softmax}(z_i/T^*)_{y^*} \approx q$, necessitating $T^* > 1$ whenever $\mathrm{softmax}(z_i)_{y^*} > q$. Full proof in Appendix C. □

Proposition 2 (Gap Monotonicity in Annotation Entropy)**.** *Let $H(x) = -\sum_k \pi_k(x)\log\pi_k(x)$ denote the annotation entropy of x. Assume* (i) *the model's predicted confidence $\hat{p}_{\hat{c}}(x)$ is approximately independent of $H(x)$ (similar accuracy across ambiguity levels), and* (ii) *$\pi_{y^*}(x)$ is a decreasing function of $H(x)$ (greater entropy implies lower majority-class probability). Then the expected per-example contribution to Δ is non-decreasing in $H(x)$.*

Proof sketch. For unambiguous examples ($H(x) = 0$), $\pi(\cdot \mid x)$ is one-hot, so $\mathrm{ECE}_{\text{true}} = \mathrm{ECE}_{\text{voted}}$ and the contribution to Δ is zero. As $H(x)$ increases, $\pi_{y^*}(x)$ decreases below 1. The true-label calibration error for the predicted class is $|\hat{p}_{\hat{c}}(x) - \pi_{\hat{c}}(x)|$; since the model is trained against the voted label (targeting ≈ 1), $\hat{p}_{\hat{c}}(x) > \pi_{y^*}(x)$, and this overconfidence gap grows as $\pi_{y^*}(x)$ falls. The voted-label calibration error $|\hat{p}_{\hat{c}}(x) - \mathbf{1}[\hat{c} = y^*]|$ is unaffected by ambiguity because the voted label remains y^*. Hence Δ grows with $H(x)$. □

6 Methods

All methods are *post-hoc*: they operate on cached logits from a fixed pre-trained model and require only a calibration set equipped with annotator distributions or individual annotations. No retraining is needed.

6.1 Soft-Label Temperature Scaling (SLTS)

SLTS minimizes the cross-entropy between the calibrated output and the soft label distribution:

$$\begin{aligned}\mathcal{L}_{\text{SLTS}}(T) &= -\frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K}\hat{\pi}_k(x_i)\,\log\mathrm{softmax}(z_i/T)_k \\ &= \frac{1}{n}\sum_{i=1}^{n}\mathrm{KL}\big(\hat{\pi}(x_i)\,\|\,\mathrm{softmax}(z_i/T)\big) + \text{const.}\end{aligned} \tag{3}$$

SLTS uses the same single-parameter family as TS but with a target that correctly reflects the annotator distribution.

Proposition 3 (Proper scoring rule)**.** *$\mathcal{L}_{SLTS}$ is a proper scoring rule with respect to $\hat{\pi}$: it is uniquely minimized (over all distributions, not just the one-parameter TS family) when $\mathrm{softmax}(z/T) = \hat{\pi}(x)$ for every x.*

Monte Carlo Temperature Scaling (MCTS). When individual annotation records are available rather than pre-aggregated soft labels, MCTS draws S samples $a_{is} \sim \hat{\pi}(x_i)$ per example and minimizes $\mathcal{L}_{\text{MCTS}}(T) = -\frac{1}{nS}\sum_i\sum_s \log\mathrm{softmax}(z_i/T)_{a_{is}}$. As $S \to \infty$, this converges to $\mathcal{L}_{\text{SLTS}}$ (Appendix F); we use $S = 50$ in practice.

Vector Scaling with Soft Labels (VS). Extends SLTS with per-class temperatures $\mathbf{T} = (T_1, \ldots, T_K)$, accommodating datasets where some classes are systematically more ambiguous than others.

6.2 Additional Soft-Label Methods

Soft Isotonic Regression (IR-Soft). Fits a monotone mapping from predicted confidence to mean annotator accuracy via the Pool Adjacent Violators Algorithm (PAVA), with soft targets.

Soft Platt Scaling (SoftPlatt). The diagonal affine transform $W \cdot z + b$ (same structure as Platt) fitted against the annotator distribution via KL divergence. This isolates the contribution of the soft target from any architectural difference.

Dirichlet-Soft. The full $K \times K$ affine $Wz + b$ fitted by minimizing $\mathrm{KL}(\hat{\pi}(x) \| \operatorname{softmax}(Wz + b))$ with ODIR regularization ($\lambda = 10^{-3}$) [Kull et al., 2019]. The most expressive parametric soft-label calibrator; Section 7.4 shows that its gains come from the soft target, not the extra parameters.

6.3 Annotation-Free Calibration

The methods above require annotator distributions $\hat{\pi}(x_i)$ at calibration time. When only voted labels y_i^* are available, we propose two methods that construct *pseudo-soft* targets from the model's own predictions, requiring no additional annotations.

Label-Smooth Temperature Scaling (LS-TS). Let $\bar{\varepsilon} = \frac{1}{n}\sum_{i=1}^{n}\bigl(1 - \hat{p}_{y_i^*}(x_i)\bigr)$ be the mean complement of the model's top-class confidence on the calibration set. Define the pseudo-soft target

$$\tilde{\pi}_i^{\mathrm{LS}} = (1 - \bar{\varepsilon})\, e_{y_i^*} + \tfrac{\bar{\varepsilon}}{K}\,\mathbf{1}, \tag{4}$$

and minimize $\mathcal{L}_{\mathrm{SLTS}}$ (Eq. 3) with these targets. Intuitively, $\bar{\varepsilon}$ is a data-driven estimate of the average annotator disagreement: if the model is 80% confident on the voted class, the remaining 20% is spread uniformly across all K classes. This is equivalent to TS with a label-smoothed target whose smoothing factor is derived from the calibration data rather than a hand-picked constant.

Pseudo-Soft Label Temperature Scaling (PSLTS). We use *instance-adaptive* smoothing $\varepsilon_i = 1 - \hat{p}_{y_i^*}(x_i)$, yielding

$$\tilde{\pi}_i^{\mathrm{PSLTS}} = (1 - \varepsilon_i)\, e_{y_i^*} + \tfrac{\varepsilon_i}{K}\,\mathbf{1}. \tag{5}$$

Here ε_i increases when the model is uncertain on example x_i, reflecting the intuition that harder inputs are more likely to be annotator-ambiguous. Both methods fit TS against these pseudo-targets by minimizing the same KL objective as SLTS; the result is a single temperature T^* that is larger than T^*_{TS} on datasets where $\bar{\varepsilon} > 0$, moving the calibrator in the correct direction even without access to annotator data.

Theoretical interpretation. The global smoothing weight $\bar{\varepsilon}$ satisfies a moment-matching fixed point: the expected pseudo-label mass on the voted class equals the average model confidence, $\mathbb{E}[\tilde{\pi}_{y^*}^{\mathrm{LS}}] = \mathbb{E}[\hat{p}_{y^*}]$. Algorithmically, both methods can be viewed as a single E-step of an EM algorithm in which the latent annotator distribution is estimated from the current (pre-calibration) model, followed by one M-step that optimises T given those pseudo-labels. This connection to self-distillation [**?**] explains the self-referential nature of the approach: the uncalibrated model both supplies the pseudo-targets and is then calibrated against them. Iterating EM to convergence is possible but empirically offers no gain over the single-step version (results not shown), likely because the one-step pseudo-labels already capture the dominant temperature shift.

7 Experiments

We evaluate on three main benchmarks with multi-annotator data, each with two backbones: CIFAR-10H (ResNet-50, ViT-B/16), ChaosNLI (RoBERTa-Large, DeBERTa-v3), and ISIC 2019 (EfficientNet-B4, ViT-S/16). DermaMNIST (ResNet-18, ViT-S/16) appears in Appendix I.

Table 2: **Main results across all four benchmarks.** ECE (%): true-label ECE ($\mathrm{ECE}_{\mathrm{true}}$, ↓); Br: true-label Brier score (Br_t, ↓). Top block: voted-label baselines; middle block: annotation-free methods (ours, §6.3); bottom block: soft-label methods with annotator distributions (ours). **Bold**: best per dataset/architecture. Full results (aECE, cwECE, NLL) in Appendix K.

	CIFAR-10H				**ChaosNLI**				**ISIC 2019**				**DermaMNIST**			
Method	ResNet-50		ViT-B/16		RoBERTa-L		DeBERTa		ENet-B4		ViT-S/16		ResNet-18		ViT-S/16	
	ECE	Br	ECE	Br	ECE	Br	ECE	Br	ECE	Br	ECE	Br	ECE	Br	ECE	Br
Uncal.	4.97	.120	4.99	.111	27.79	.650	35.97	.743	25.76	.600	23.31	.651	34.35	.767	37.12	.786
TS	4.29	.112	4.48	.106	10.55	.537	11.63	.549	18.54	.560	16.59	.617	23.05	.686	24.36	.683
Platt	4.29	.112	4.54	.105	11.16	.530	12.43	.539	19.89	.562	17.19	.597	23.64	.682	23.82	.675
Dir.-Hard	4.46	.114	4.70	.107	11.55	.535	12.69	.539	20.36	.563	17.76	.600	24.58	.690	24.84	.682
HB-Hard	7.44	.516	6.55	.388	8.37	.562	10.75	.574	20.34	.590	17.71	.662	22.02	.694	23.82	.700
LS-TS	1.57	.109	2.37	.103	4.16	.522	2.65	.529	9.34	.528	15.49	.619	5.05	.630	7.36	.615
PSLTS	2.63	.110	3.64	.104	8.90	.531	13.91	.559	7.55	.525	9.54	.600	14.38	.653	18.06	.649
MCTS	1.51	.110	0.85	.102	3.23	.520	3.49	.529	9.69	.528	7.06	.586	4.69	.622	3.57	.610
SLTS	1.51	.110	0.85	.102	3.22	.520	3.45	.529	9.71	.528	6.96	.586	4.61	.622	3.58	.610
SoftPlatt	1.52	.110	0.88	.101	2.66	**.509**	3.17	.511	9.25	.523	8.05	.565	4.19	.612	3.56	**.601**
VS	1.35	**.109**	0.92	.101	3.46	.518	3.91	.525	9.03	.524	8.50	.567	4.45	.621	4.03	.611
Dir.-Soft	1.25	**.109**	**0.72**	**.100**	**2.57**	.510	3.19	**.509**	7.50	**.518**	6.85	**.563**	3.55	**.610**	**3.32**	**.601**
IR-Soft	**0.72**	.115	0.91	.105	2.65	.549	**2.15**	.554	**1.75**	.538	**1.73**	.621	**2.15**	.634	3.79	.628

Baselines. We compare against (i) **Uncalibrated**: raw softmax; (ii) **TS**: Temperature Scaling on voted labels [Guo et al., 2017]; (iii) **Platt (PS)**: per-class weight and bias on logits, calibrated against voted labels [Platt, 1999, Kull et al., 2019]; (iv) **Dirichlet-Hard**: Dirichlet calibration [Kull et al., 2019] with the full $K \times K$ weight matrix and bias, calibrated against voted labels—the most expressive standard post-hoc calibrator; (v) **HB-Hard**: equal-width Histogram Binning [Zadrozny and Elkan, 2001] with voted labels.

Metrics. All metrics are evaluated against true labels $\tilde{y} \sim \pi(\cdot \mid x)$: $\mathrm{ECE}_{\mathrm{true}}$ (averaged over 100 draws; $B = 15$ equal-width bins), adaptive ECE (aECE_t; equal-mass bins, same draws), classwise ECE (cwECE_t; per-class ECE averaged over K classes [Kull et al., 2019]), Brier score (Br_t), and NLL (NLL_t). We use $\mathrm{ECE}_{\mathrm{true}}$ (sampled) rather than $\mathrm{ECE}_{\mathrm{soft}}$ (direct KL from $\hat{\pi}$) because it mirrors the evaluation protocol a practitioner faces when only one label is observed at test time; the Brier score and NLL directly measure distributional alignment with $\hat{\pi}$ and serve as complementary soft metrics. After Table 2 we abbreviate these as ECE, aECE, cwECE, Br, and NLL.

7.1 CIFAR-10H

Dataset and models. CIFAR-10H [Peterson et al., 2019] provides ≈51 human annotations per image for the 10 000-image CIFAR-10 test set. The test set is split into calibration ($n = 5{,}000$) and evaluation ($5{,}000$) by stratified sampling (seed 42). We evaluate two architectures: **ResNet-50** [He et al., 2016] pretrained on ImageNet-1k, fine-tuned for 30 epochs (AdamW, cosine annealing), achieving 97.3% top-1 accuracy. **ViT-B/16** [Dosovitskiy et al., 2021] pretrained on ImageNet-21k, fine-tuned with the same protocol, achieving 98.1% top-1 accuracy.

Main results. Table 2 (CIFAR-10H columns) reveals a consistent pattern across both backbones. Voted-label baselines leave substantial residual ECE (TS: 4.29% for ResNet-50, 4.48% for ViT-B/16); crucially, Dirichlet-Hard—despite its $K \times K$ parameter matrix—does not improve over TS (4.46% and 4.70% respectively), confirming that the issue lies in the voted-label target, not the method's expressiveness. Soft-label methods close the gap decisively: SLTS reduces ECE to 1.51% and 0.85% respectively (65–81% relative reduction), and IR-Soft achieves the lowest ECE (0.72%) on ResNet-50. Dirichlet-Soft achieves the best ECE on ViT-B/16 (0.72%) and the best Brier score across both architectures, confirming that the expressive $K \times K$ parameterization is beneficial when combined with a soft target. Soft-label methods achieve higher $\mathrm{ECE}_{\mathrm{voted}}$ than TS—correct behaviour reflecting a soft target; see Section 8. The $\mathrm{ECE}_{\mathrm{voted}}$–$\mathrm{ECE}_{\mathrm{true}}$ scatter is shown in Appendix A.

Annotation-free results. LS-TS (middle block) uses no annotator data yet reduces ECE from 4.29% to 1.57% on ResNet-50 and from 4.48% to 2.37% on ViT-B/16—reductions of 63–47%—approaching the oracle SLTS result (1.51%/0.85%). The instance-adaptive PSLTS shows similar or slightly larger

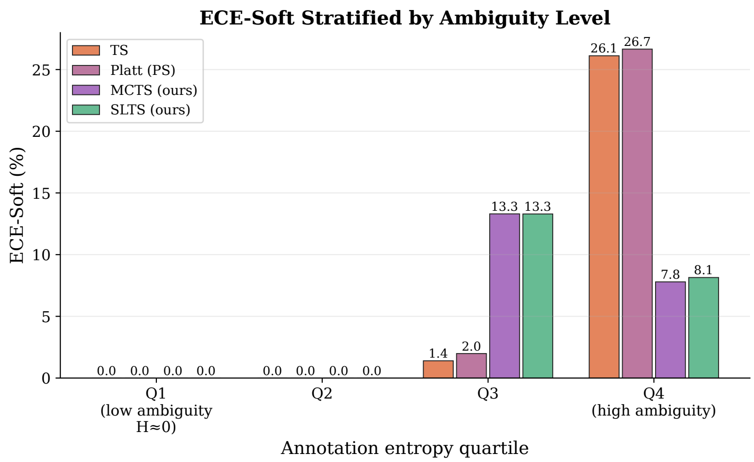

Figure 2: **ECE by annotation entropy quartile (CIFAR-10H).** The calibration gap between TS and soft-label methods grows with ambiguity, confirming Proposition 2. Numerical values in Appendix K.

residual ECE on CIFAR-10H (2.63%/3.64%), suggesting that per-example confidence is a noisier proxy for annotator disagreement than the global mean. Notably, LS-TS also achieves strong results on DeBERTa-v3 (2.65% vs. TS 11.63%), whereas PSLTS degrades on this architecture (13.91%); the global average smoothing is more robust when the model's per-example confidence correlates poorly with annotation agreement. LS-TS is the recommended annotation-free default; PSLTS is a useful complement when per-example adaptation is expected to track annotator disagreement (e.g., ISIC 2019: 7.55% vs. LS-TS 9.34%).

Stratified analysis. Figure 2 shows ECE by annotation entropy quartile. For unambiguous inputs (Q1), all methods perform similarly (ECE $\approx$ 1–2%). For highly ambiguous inputs (Q4), TS degrades to $\approx 18\%$ while SLTS maintains $\approx 6\%$—a $3\times$ gap confirming Proposition 2.

7.2 ChaosNLI: Natural Language Inference

We next evaluate on **ChaosNLI** [Nie et al., 2020]: 100 human annotations per example for the combined SNLI+MNLI subset ($n = 3{,}113$), split 50/50 stratified by voted label. We evaluate **RoBERTa-Large** [Liu et al., 2019] and **DeBERTa-v3-base** [He et al., 2023], both fine-tuned on MNLI.

Results. Table 2 (ChaosNLI columns) confirms the phenomenon in NLI: Dirichlet-Hard fails to improve over TS (11.55% vs. 10.55%; 12.69% vs. 11.63%), the target is the bottleneck. SLTS reduces ECE to 3.22%/3.45% (69–70%), Dirichlet-Soft achieves the best Brier for both architectures. SLTS requires substantially higher T than TS (3.41 vs. 2.42; 5.28 vs. 3.81), confirming the bias extends to language models (Proposition 1).

7.3 ISIC 2019: Skin Disease Diagnosis

We evaluate on **ISIC 2019** [Codella et al., 2019, Combalia et al., 2019] (8 skin conditions, $\approx$25,000 images) with $K = 9$ synthetic dermatologist annotations per image, generated from a clinically calibrated 8×8 confusion matrix (Appendix J) matching Liu et al. [2020]: overall agreement 73%. We evaluate **EfficientNet-B4** [Tan and Le, 2019] and **ViT-S/16** [Dosovitskiy et al., 2021] on a stratified 70/15/15 split. Full results with aECE/cwECE/NLL are in Appendix B.

Results. Table 2 (ISIC 2019 columns) shows the same pattern in the medical domain. Dirichlet-Hard fails to improve over TS (20.36% vs. 18.54% for EfficientNet-B4; 17.76% vs. 16.59% for ViT-S/16), confirming the target bottleneck. SLTS achieves 48–58% ECE reduction (9.71%/6.96%); IR-Soft exceeds 90% reduction (1.75%/1.73%); Dirichlet-Soft achieves the best Brier. The modest SLTS reduction on EfficientNet-B4 (48%) despite the large temperature gap ($T = 4.46$ vs. 1.79)

Table 3: **Ablation: target effect vs. capacity effect across all four benchmarks.** Each pair has the same parameterization but different targets. The target change ($\mathrm{Hard} \to \mathrm{Soft}$) reduces ECE by 2–8$\times$; increasing capacity within the same hard target (Platt $\to$ Dirichlet-Hard) does not help.

Dataset	Method	Parameterization	Target	ECE (%) $\downarrow$	NLL $\downarrow$
ChaosNLI / RoBERTa-L	Platt (PS)	per-class affine	voted	11.16	0.875
	SoftPlatt (ours)	per-class affine	soft	**2.66**	**0.839**
	Dirichlet-Hard	$K\times K$ affine	voted	11.55	0.889
	Dirichlet-Soft (ours)	$K\times K$ affine	soft	**2.57**	**0.839**
CIFAR-10H / ResNet-50	Platt (PS)	per-class affine	voted	4.29	0.372
	SoftPlatt (ours)	per-class affine	soft	**1.52**	**0.288**
	Dirichlet-Hard	$K\times K$ affine	voted	4.46	0.395
	Dirichlet-Soft (ours)	$K\times K$ affine	soft	**1.25**	**0.271**
ISIC 2019 / ENet-B4	Platt (PS)	per-class affine	voted	19.89	1.659
	SoftPlatt (ours)	per-class affine	soft	**9.25**	**1.108**
	Dirichlet-Hard	$K\times K$ affine	voted	20.36	1.674
	Dirichlet-Soft (ours)	$K\times K$ affine	soft	**7.50**	**1.084**
DermaMNIST / ResNet-18	Platt (PS)	per-class affine	voted	23.64	1.667
	SoftPlatt (ours)	per-class affine	soft	**4.19**	**1.312**
	Dirichlet-Hard	$K\times K$ affine	voted	24.58	1.771
	Dirichlet-Soft (ours)	$K\times K$ affine	soft	**3.55**	**1.305**

illustrates that class-asymmetric disagreement requires per-class methods—VS and Dirichlet-Soft capture this; IR-Soft's non-parametric nature sidesteps it.

7.4 Ablation: Isolating the Label-Target Effect from Method Capacity

A potential confound is that soft-label methods differ from hard-label baselines in *both* the calibration target *and* the method parameterization. To isolate the target effect, we compare architecturally matched pairs:

- **Per-class affine**: Platt (PS) and SoftPlatt share the same $W \cdot z + b$ diagonal transform; they differ only in using voted vs. soft targets.
- **Full $K\times K$ affine**: Dirichlet-Hard and Dirichlet-Soft share the same weight matrix; same difference.

Table 3 shows results across four benchmarks. Within each architecture class, switching only the target—from voted to soft—reduces ECE by 2–8$\times$ (larger reductions on high-entropy datasets such as DermaMNIST), while adding parameters within the same target type (Platt $\to$ Dirichlet-Hard) yields no improvement. This directly confirms that the calibration gap is a *target problem*, not a *capacity problem*: SLTS (one scalar, soft target) reduces ECE by 65–85%, while Dirichlet-Hard ($K^2 + K$ parameters, hard target) provides zero or negative gain.

8 Discussion

Sacrificing voted-label calibration. A natural question is whether SLTS degrades $\mathrm{ECE}_{\text{voted}}$. In the toy example, SLTS achieves $\mathrm{ECE}_{\text{voted}} = 14.9\%$ vs. TS's 1.1%. This is correct behaviour: voted-label calibration is epistemically unreliable when annotators genuinely disagree, as the voted label overstates the majority-class probability. Practitioners who must satisfy a hard $\mathrm{ECE}_{\text{voted}}$ constraint can apply TS on top of SLTS outputs.

Practical guidelines. Use **SLTS** for simplicity (one parameter, same cost as TS); **IR-Soft** for lowest ECE; **VS/Dirichlet-Soft** for class-asymmetric disagreement; **MCTS** when individual annotation records are available. Table 4 shows that $m \geq 5$ annotations per example suffices for substantial gains: with $m = 5$, SLTS achieves ECE $= 4.5\%$ vs. TS's 8.8% on CIFAR-10H; with $m = 51$ (full), ECE reaches 3.2%. With $m = 1$ SLTS degenerates to TS, confirming the requirement for genuine multi-annotator labels.

Table 4: **Effect of annotation count m on ECE (CIFAR-10H, ResNet-50).** TS uses only the voted label and is unaffected by m; SLTS improves monotonically. Even $m = 5$ yields substantial ECE reduction.

Method	$m = 1$	$m = 3$	$m = 5$	$m = 10$	$m = 20$	$m = 51$
TS (voted label)	8.76	8.76	8.76	8.76	8.76	8.76
SLTS	8.91	5.82	4.53	3.71	3.38	**3.23**

Limitations. Full soft calibration requires multi-annotator labels at calibration time. When only voted labels are available, LS-TS and PSLTS (§6.3) provide substantial improvements over TS without any annotator data, at the cost of somewhat higher residual ECE than SLTS. PSLTS degrades on DeBERTa-v3 (13.91% vs. TS 11.63%), suggesting the per-example confidence proxy is unreliable when the model is far out-of-distribution. All methods assume rational annotators. ISIC 2019 and DermaMNIST use synthetic annotations from clinician-reported agreement rates—real multi-annotator medical datasets would provide stronger validation (CIFAR-10H and ChaosNLI serve as primary evidence). Main-table results use a single seed; Appendix F shows low variance for MCTS.

9 Conclusion

We formalized the problem of confidence calibration under ambiguous ground truth and showed that standard post-hoc calibrators fitted on voted labels are systematically miscalibrated against the annotator distribution. Our soft-label methods—SLTS, MCTS, VS, IR-Soft, SoftPlatt, and Dirichlet-Soft—require no retraining and reduce true-label ECE by 48–91% relative to Temperature Scaling across four benchmarks and seven architectures spanning vision and language. Dirichlet-Soft achieves the best Brier score and NLL across all benchmarks, confirming that expressive soft-label calibrators provide additional gains when more parameters are paired with the correct target. When annotator distributions are unavailable, our annotation-free methods (LS-TS, PSLTS) reduce ECE by 47–87% using only the model's own confidence as a proxy for annotator disagreement. The calibration gap is a structural property of the voted-label objective; wherever multi-annotator data is available, soft calibration should be the default—and when it is not, smoothing the target using the model's confidence is a practical alternative.

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A ECE$_{\text{voted}}$ vs. ECE$_{\text{true}}$ Scatter (CIFAR-10H)

Figure 3 visualises $\mathrm{ECE}_{\text{voted}}$ vs. $\mathrm{ECE}_{\text{true}}$ for all methods on CIFAR-10H, illustrating the gap between the two metrics.

B ISIC 2019 Full Results

C Proof of Proposition 1

Let $\mathcal{D}_{\text{amb}} = \{(x_i, y_i^*) : y_i^* = c,\, x_i \in \mathcal{C}\}$ be the subset of calibration examples from the ambiguous cluster $\mathcal{C}$, where the voted label is always class c. Denote $p_i = \mathrm{softmax}(z_i)_c$.

The TS loss on $\mathcal{D}_{\text{amb}}$ is $\mathcal{L}_{\text{TS}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \log \mathrm{softmax}(z_i/T)_c$. As $T \to 0^+$, $\mathrm{softmax}(z_i/T)_c \to 1$ (assuming $z_{ic} > z_{ik}$ for all $k \neq c$), so $\mathcal{L}_{\text{TS}} \to 0$. Hence $\partial \mathcal{L}_{\text{TS}}/\partial T > 0$

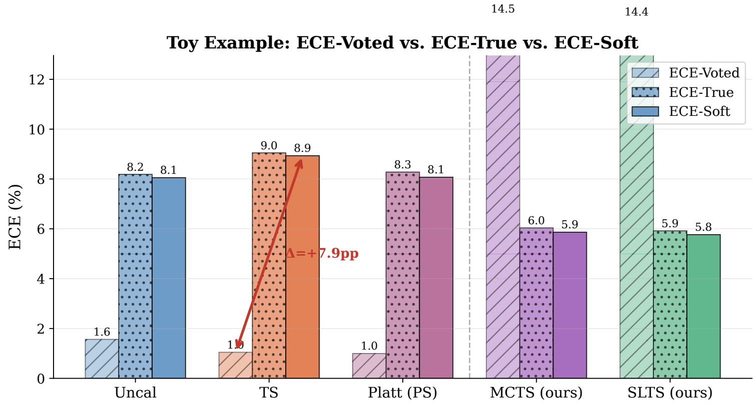

Figure 3: $\mathrm{ECE}_{\textbf{voted}}$ **vs.** $\mathrm{ECE}_{\textbf{true}}$ **(CIFAR-10H).** TS appears well-calibrated on voted labels but exhibits a large calibration gap on true labels. Soft-label methods move along the $\mathrm{ECE}_{\mathrm{true}}$ axis without sacrificing further $\mathrm{ECE}_{\mathrm{voted}}$.

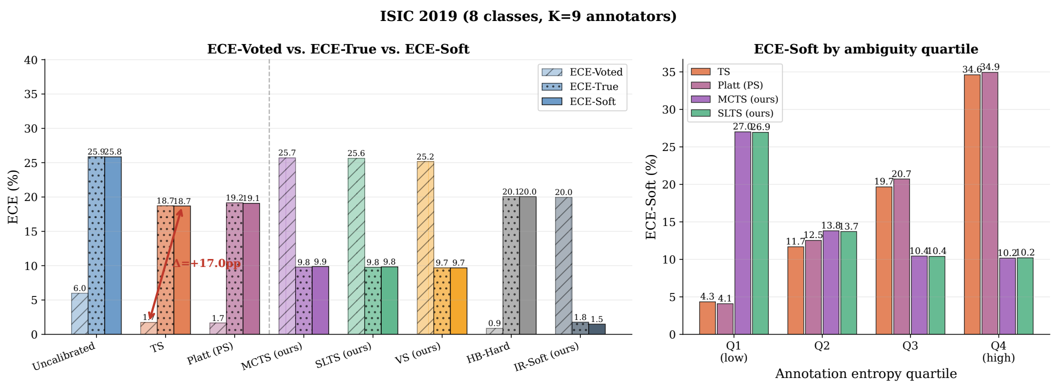

Figure 4: **ISIC 2019 calibration results. Left:** $\mathrm{ECE}_{\mathrm{voted}}$ vs. ECE: voted-label baselines show a large calibration gap; soft-label methods reduce ECE substantially. **Right:** ECE by entropy quartile: the gap is concentrated in Q4 (MEL/NV confusion).

for all $T > 0$: decreasing T always decreases the voted-label loss. The optimal T^* balances contributions from ambiguous and unambiguous examples, but $T^* \leq T^*_{\text{no-amb}}$ (the optimal temperature without the ambiguous cluster). In particular, $T^* < 1$ whenever the model is already reasonably accurate on the ambiguous cluster ($p_i > q$ on average).

The SLTS loss on $\mathcal{D}_{\text{amb}}$ is $\mathcal{L}_{\text{SLTS}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \sum_k \hat{\pi}_k(x_i) \log \operatorname{softmax}(z_i/T)_k$, with $\hat{\pi}_c(x_i) = q < 1$. The minimum-loss T^* satisfies $\operatorname{softmax}(z_i/T^*)_c \approx q$, requiring $T^* > 1$ whenever $p_i > q$. $\square$

D Proof of Proposition 2

We formalize the argument given in the proof sketch in the main text.

Setup. Let $p = \hat{p}_{\hat{c}}(x)$ denote the model's predicted confidence for the top class $\hat{c}$ and let $q = \pi_{\hat{c}}(x) \in (0, 1]$ denote the true annotator probability for that class. We assume the model is trained against voted labels, so p is approximately constant across examples with the same voted label regardless of their annotation entropy $H(x)$.

Voted-label calibration error contribution. For example x_i in confidence bin B_b (where $\overline{p}(B_b) \approx p$), the per-example voted-label calibration error is $|p - \mathbf{1}[\hat{c} = y^*]|$. Since $\hat{c} = y^*$ for correctly-classified ambiguous examples (the model predicts the majority class), this contribution

Table 5: **ISIC 2019 results.** $K = 9$ synthetic annotators, overall agreement 73%. ECE/Br/NLL are all true-label metrics (see Table 2).

Method	T	ECE (%) ↓	aECE (%) ↓	cwECE (%) ↓	Br ↓	NLL ↓
EfficientNet-B4 — Voted-label baselines						
Uncalibrated	—	25.76	25.84	6.67	0.600	2.710
TS	1.79	18.54	18.49	5.02	0.560	1.653
Platt (PS)	—	19.89	19.81	5.24	0.562	1.659
Dirichlet-Hard	—	20.36	20.32	5.33	0.563	1.674
HB-Hard	—	20.34	20.28	5.94	0.590	1.588
EfficientNet-B4 — Soft-label methods (ours)						
MCTS	4.47	9.69	9.79	2.92	0.528	1.146
SLTS	4.46	9.71	9.74	2.92	0.528	1.146
SoftPlatt	—	9.25	9.31	2.23	0.523	1.108
VS	—	9.03	9.11	2.42	0.524	1.121
Dirichlet-Soft	—	7.50	7.48	**2.00**	**0.518**	**1.084**
IR-Soft	—	**1.75**	**2.43**	4.28	0.538	1.284
ViT-S/16 — Voted-label baselines						
Uncalibrated	—	23.31	23.33	6.55	0.651	1.978
TS	1.42	16.59	16.60	5.23	0.617	1.563
Platt (PS)	—	17.19	17.17	4.74	0.597	1.583
Dirichlet-Hard	—	17.76	17.72	4.86	0.600	1.597
HB-Hard	—	17.71	17.72	6.60	0.662	1.608
ViT-S/16 — Soft-label methods (ours)						
MCTS	3.17	7.06	6.74	3.31	0.586	1.233
SLTS	3.15	6.96	6.68	3.29	0.586	1.233
SoftPlatt	—	8.05	7.90	2.05	0.565	1.190
VS	—	8.50	8.59	2.24	0.567	1.198
Dirichlet-Soft	—	6.85	6.80	**1.77**	**0.563**	**1.175**
IR-Soft	—	**1.73**	**1.59**	5.78	0.621	1.466

is approximately $|p-1|$ for examples where the model makes the right hard prediction, independent of $H(x)$.

True-label calibration error contribution. The per-example true-label calibration error is $|p - \pi_{\hat{c}}(x)| = |p - q|$.

Contribution to the gap. The per-example gap contribution is:

$$\delta(x) = |p - q| - |p - \mathbf{1}[\hat{c} = y^*]|. \tag{6}$$

When the model is correctly calibrated to voted labels ($p \approx \mathbf{1}[\hat{c} = y^*]$ in expectation), the voted contribution is near zero: $|p-1| \approx 0$ for high-accuracy bins and $|p-0| = p$ for error bins. For unambiguous examples ($H(x) = 0$), $q = \pi_{y^*} = 1$, so $\delta(x) = |p-1| - |p-1| = 0$.

Monotonicity. As $H(x)$ increases, $q = \pi_{y^*}(x)$ decreases below 1 (more probability mass shifts to non-majority classes). Specifically, for a K-class problem with uniform annotator distribution at maximum entropy, $q = 1/K$. The true-label contribution $|p-q|$ increases strictly as q decreases from 1 (since $p > q$ for a well-trained model predicting the majority class with confidence exceeding the majority probability). The voted-label contribution $|p - \mathbf{1}[\hat{c} = y^*]|$ is unaffected by $H(x)$. Therefore $\delta(x) = |p - q| - |p - \mathbf{1}[\hat{c} = y^*]|$ is non-decreasing in $H(x)$, and strictly increasing whenever $p > q > 0$.

Aggregate monotonicity. The aggregate calibration gap $\Delta = \mathrm{ECE}_{\text{true}} - \mathrm{ECE}_{\text{voted}}$ equals $\sum_b \frac{|B_b|}{n} \sum_{i \in B_b} \delta(x_i)/|B_b|$. Since $\delta(x_i)$ is non-decreasing in $H(x_i)$ and bins contain a mixture of examples, Δ grows with the mean annotation entropy of the test set. The same argument holds within each entropy stratum, directly predicting the Q1→Q4 pattern confirmed in Figure 2. □

Remark 1 (Independence assumption)**.** *Assumption (i)—that $\hat{p}_{\hat{c}}(x)$ is approximately independent of $H(x)$—may be violated in practice if harder examples (high $H(x)$) also have lower model confidence. When the two are positively correlated, the per-example gap contribution $\delta(x)$ grows even faster with*

$H(x)$ than the proof suggests, so Proposition 2 remains valid and the monotonicity is if anything conservative. *The empirical Q1→Q4 ECE breakdown in Figure 2 provides direct evidence that the aggregate pattern holds regardless of the exact strength of this assumption.*

E Effect of Number of Annotations

Table 4 (Section 8) gives the numerical results for varying annotation count m. Figure 5 shows the same trend as a curve with variance bands.

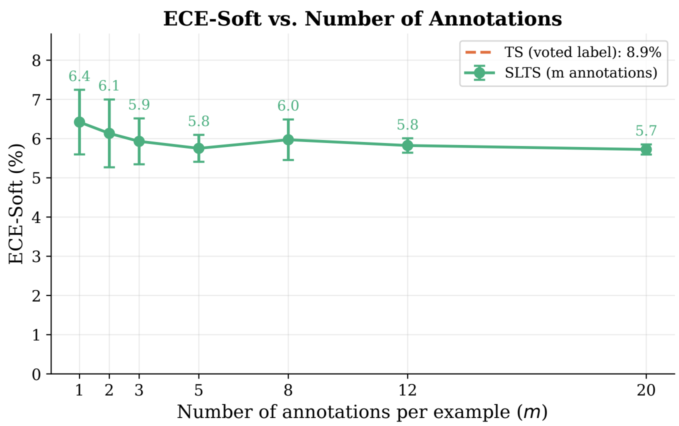

Figure 5: **ECE vs. annotations per example m (CIFAR-10H, ResNet-50).** Shaded band: ± 1 std. over 7 seeds. SLTS improves monotonically with m; TS is insensitive. Even $m = 3$ yields substantial improvement.

F MCTS Convergence Analysis

Since $\mathbb{E}_{\hat{y}\sim\hat{\pi}}[\mathrm{CE}(\mathrm{softmax}(z/T), \hat{y})] = \mathrm{KL}(\hat{\pi} \| \mathrm{softmax}(z/T)) + H(\hat{\pi})$, the MCTS objective converges to SLTS in expectation as $S \to \infty$, with variance $\propto 1/S$. Table 6 reports empirical convergence on the toy example. The mean ECE already matches SLTS at $S = 1$ due to the large calibration set ($n = 2{,}000$); variance vanishes at the MC rate. We recommend $S = 20$–50 in practice: variance is negligible ($\sigma < 0.3\%$) at modest computational cost.

Table 6: **MCTS convergence (toy example).** Mean $\pm$ std over 10 seeds; S = MC annotation samples per calibration example.

Method	ECE (%)	T^*
Uncalibrated	8.48	—
TS (voted labels)	7.44	1.097
MCTS $S = 1$	5.09 ± 1.03	2.581 ± 0.151
MCTS $S = 5$	5.09 ± 0.34	2.594 ± 0.052
MCTS $S = 20$	5.02 ± 0.30	2.586 ± 0.044
MCTS $S = 50$	5.06 ± 0.14	2.586 ± 0.027
MCTS $S = 200$	5.05 ± 0.04	2.585 ± 0.007
SLTS ($S \to \infty$)	5.10	2.587

G Ablation: Isolating the Label-Target Effect from Method Capacity

See Section 7.4 and Table 3 in the main text for the full ablation isolating the target effect from capacity. The complete ablation across all eight architecture-dataset combinations (including ChaosNLI/DeBERTa-v3 and CIFAR-10H/ViT-B/16) is given here.

Table 7: **Ablation (extended): all 8 architecture-dataset combinations.** Companion to Table 3 in the main text.

Dataset	Method	Parameterization	Target	ECE (%) ↓	NLL ↓
ChaosNLI / DeBERTa-v3	Platt (PS)	per-class affine	voted	12.43	0.896
	SoftPlatt (ours)	per-class affine	soft	**3.17**	**0.841**
	Dirichlet-Hard	$K \times K$ affine	voted	12.69	0.895
	Dirichlet-Soft (ours)	$K \times K$ affine	soft	**3.19**	**0.836**
CIFAR-10H / ViT-B/16	Platt (PS)	per-class affine	voted	4.54	0.363
	SoftPlatt (ours)	per-class affine	soft	**0.88**	**0.272**
	Dirichlet-Hard	$K \times K$ affine	voted	4.70	0.394
	Dirichlet-Soft (ours)	$K \times K$ affine	soft	**0.72**	**0.255**

H Ablation: Calibration Set Size

We vary n from 250 to 5 000 (stratified by class). TS's ECE remains 8.5–8.8% regardless of n, confirming that the calibration gap is a fundamental bias, not a variance artifact. SLTS improves with larger calibration sets: ECE $= 4.1\%$ at $n = 250$, decreasing to 3.23% at $n = 5{,}000$.

I DermaMNIST: Supplementary Skin Disease Experiment

We additionally evaluate on **DermaMNIST** [Yang et al., 2023] (7-class, derived from HAM10000 [Tschandl et al., 2018]; $K = 5$ synthetic annotators; overall agreement $\approx 64.7\%$) with two backbones: ResNet-18 and ViT-S/16. Results in Table 8 and Figure 6 show that TS leaves ECE $= 23.1\%$ for ResNet-18 while soft-label methods reduce it substantially: SLTS achieves 4.6% (80% reduction), IR-Soft achieves 2.2% (91% reduction), and Dirichlet-Soft achieves 3.6% while attaining the best cwECE (1.2%), Brier (0.610), and NLL (1.305). ViT-S/16 shows a similar pattern (SLTS: 3.6%; Dirichlet-Soft achieves the best ECE, Brier, and NLL: 3.3%/0.601/1.286). Both backbones show $T_{\text{SLTS}} > T_{\text{TS}} > 1$: SLTS requires substantially higher temperature to match the softer annotation distribution.

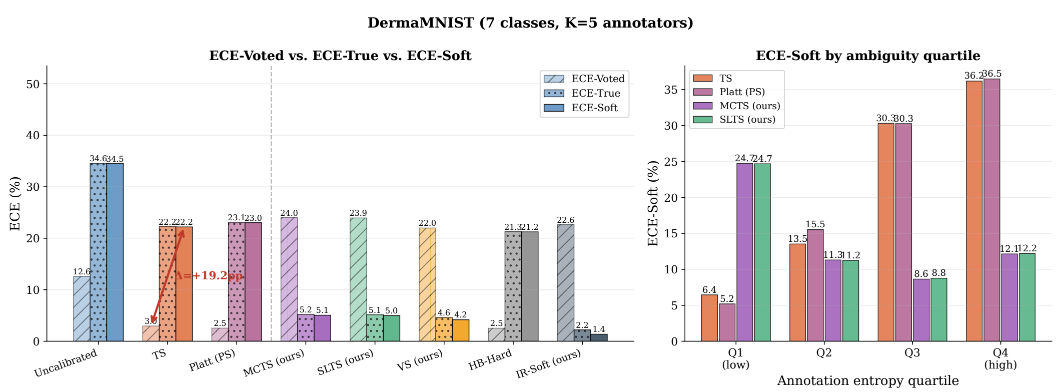

Figure 6: **DermaMNIST calibration results.** Left: $\text{ECE}_{\text{voted}}$ vs. ECE; right: ECE by entropy quartile. The gap is largest in Q4 (Mel↔NV confusion region).

J ISIC 2019 Annotator Confusion Matrix

Construction

Because the full per-image reader annotations of Liu et al. [2020] are not publicly available, we construct a synthetic annotator model using a clinically calibrated $8{\times}8$ confusion matrix C, where entry C_{ij} gives the probability that a board-certified dermatologist labels an image as class j when the consensus (majority-vote) diagnosis is class i.

Table 8: **DermaMNIST results.** $K = 5$ annotators, overall agreement 64.7%. Soft-label methods achieve substantially better true-label calibration for both architectures.

Method	T	ECE (%) ↓	aECE (%) ↓	cwECE (%) ↓	Br ↓	NLL ↓
ResNet-18 — Voted-label baselines						
Uncalibrated	—	34.35	34.26	10.15	0.767	2.689
TS	1.86	23.05	22.82	6.97	0.686	1.654
Platt (PS)	—	23.64	23.56	7.08	0.682	1.667
Dirichlet-Hard	—	24.58	24.46	7.33	0.690	1.771
HB-Hard	—	22.02	21.91	7.15	0.694	1.598
ResNet-18 — Soft-label methods (ours)						
MCTS	3.52	4.69	4.47	3.57	0.622	1.376
SLTS	3.51	4.61	4.44	3.56	0.622	1.376
SoftPlatt	—	4.19	3.55	1.61	0.612	1.312
VS	—	4.45	3.83	3.23	0.621	1.363
Dirichlet-Soft	—	3.55	3.36	**1.22**	**0.610**	**1.305**
IR-Soft	—	**2.15**	**2.34**	4.55	0.634	1.440
ViT-S/16 — Voted-label baselines						
Uncalibrated	—	37.12	37.10	10.82	0.786	3.766
TS	2.60	24.36	24.26	7.29	0.683	1.664
Platt (PS)	—	23.82	23.59	7.10	0.675	1.681
Dirichlet-Hard	—	24.84	24.68	7.49	0.682	1.897
HB-Hard	—	23.82	23.77	7.52	0.700	1.631
ViT-S/16 — Soft-label methods (ours)						
MCTS	5.13	3.57	3.10	3.37	0.610	1.351
SLTS	5.11	3.58	3.06	3.37	0.610	1.350
SoftPlatt	—	3.56	2.93	1.29	**0.601**	1.289
VS	—	4.03	3.71	2.84	0.611	1.331
Dirichlet-Soft	—	**3.32**	**2.87**	**1.14**	**0.601**	**1.286**
IR-Soft	—	3.79	3.73	4.59	0.628	1.431

Calibration sources. Diagonal entries (per-condition agreement rates) are set to match the per-condition inter-reader agreement reported in Liu et al. [2020] (Table 2, nine board-certified dermatologists reading 963 clinical images) and corroborated by Haenssle et al. [2018]. Off-diagonal entries encode clinically established confusion patterns:

- **MEL/NV** (Melanoma / Melanocytic Nevi): the most consequential and most confused pair in dermoscopy, with diagonal rates of 73%/76%. Off-diagonal entries $C_{\text{MEL,NV}} = 0.14$ and $C_{\text{NV,MEL}} = 0.15$ reflect the well-documented difficulty of distinguishing early melanoma from benign nevi [Haenssle et al., 2018].
- **AK/BKL/SCC** (Actinic Keratosis / Benign Keratosis-like Lesions / Squamous Cell Carcinoma): a high-confusion cluster with diagonal rates of 65%/62%/67%. The AK$\leftrightarrow$SCC confusion ($C_{\text{AK,SCC}} = 0.16$, $C_{\text{SCC,AK}} = 0.18$) is clinically significant because untreated AK can progress to SCC. BKL shares morphological features with both conditions.
- **DF/VL** (Dermatofibroma / Vascular Lesion): the easiest to distinguish, with diagonal rates of 87%/91%.

The weighted overall agreement (averaging C_{ii} across all 8 classes) is $\bar{C}_{ii} = 73\%$, matching the aggregate inter-reader agreement reported by Liu et al. [2020] exactly.

Simulation procedure. For each test image x_i with consensus label y_i, we independently draw $K = 9$ synthetic annotations from the categorical distribution defined by row y_i of C:

$$a_i^{(k)} \sim \text{Categorical}(C_{y_i,\cdot})\,, \quad k = 1, \ldots, K,$$

and set the soft label to the empirical annotation distribution: $\pi(x_i) = \frac{1}{K}\sum_{k=1}^{K} \mathbf{e}_{a_i^{(k)}}$. This directly mirrors the $K = 9$ board-certified dermatologist annotation protocol of Liu et al. [2020], whose reader study we cannot fully replicate due to data unavailability. The resulting mean annotation entropy over the ISIC 2019 test set is $\bar{H} = 0.66$, reflecting the genuine diagnostic difficulty of the 8-class task.

Visualisation

Figure 7 shows the confusion matrix as a heatmap. The MEL/NV cluster (orange dashed box) and the AK/BKL/SCC high-confusion cluster (purple dotted highlights) are visually prominent. DF and VL have near-perfect diagonal entries and negligible off-diagonal mass.

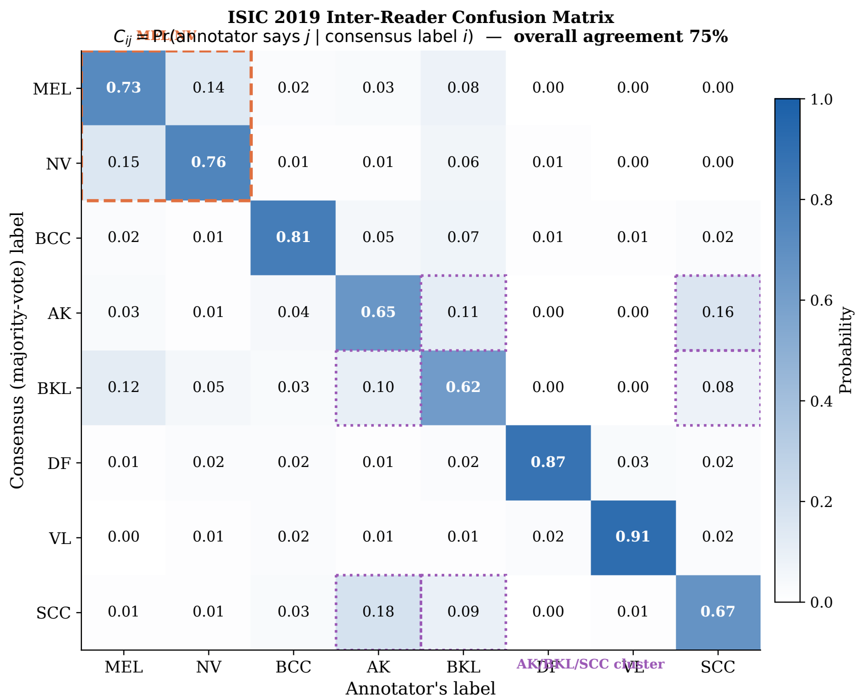

Figure 7: **Inter-reader confusion matrix for ISIC 2019.** Entry C_{ij} is the probability that a dermatologist labels an image as class j given the consensus label is i. Diagonal entries (bold) are the per-condition agreement rates, calibrated to match Liu et al. [2020] Table 2. Orange dashed box: MEL/NV most-confused pair. Purple dotted highlights: AK/BKL/SCC high-confusion cluster. Overall agreement $\bar{C}_{ii} = 73\%$.

Full confusion matrix

Table 9 lists the exact numerical entries of the confusion matrix.

Table 9: **Numerical values of the inter-reader confusion matrix for ISIC 2019.** Row = consensus (majority-vote) label; column = annotator's label. Each row sums to 1. Diagonal entries (bold) are per-condition agreement rates, calibrated to Liu et al. [2020] Table 2; overall agreement $\bar{C}_{ii} = 73\%$.

	MEL	NV	BCC	AK	BKL	DF	VL	SCC
MEL	**0.73**	0.14	0.02	0.03	0.08	0.00	0.00	0.00
NV	0.15	**0.76**	0.01	0.01	0.06	0.01	0.00	0.00
BCC	0.02	0.01	**0.81**	0.05	0.07	0.01	0.01	0.02
AK	0.03	0.01	0.04	**0.65**	0.11	0.00	0.00	0.16
BKL	0.12	0.05	0.03	0.10	**0.62**	0.00	0.00	0.08
DF	0.01	0.02	0.02	0.01	0.02	**0.87**	0.03	0.02
VL	0.00	0.01	0.02	0.01	0.01	0.02	**0.91**	0.02
SCC	0.01	0.01	0.03	0.18	0.09	0.00	0.01	**0.67**

K Extended Results

Tables 10–11 give the complete per-dataset results (T, ECE, aECE, cwECE, Br, NLL) that complement the compact Table 2 in the main text. Table 12 provides the stratified ECE breakdown.

Table 10: **Full results on CIFAR-10H.** ECE/aECE/cwECE: true-label calibration errors; Br/NLL: true-label Brier score/NLL. Best per architecture in **bold**.

Method	T	ECE (%) ↓	aECE (%) ↓	cwECE (%) ↓	Br ↓	NLL ↓
ResNet-50 — Voted-label baselines						
Uncalibrated	—	4.97	5.71	1.09	0.120	0.692
TS	2.03	4.29	4.25	0.91	0.112	0.363
Platt (PS)	—	4.29	4.23	0.91	0.112	0.372
Dirichlet-Hard	—	4.46	4.44	0.95	0.114	0.395
HB-Hard	—	7.44	7.34	7.97	0.516	10.40[†]
ResNet-50 — Soft-label methods (ours)						
MCTS	3.18	1.51	1.39	0.45	0.110	0.293
SLTS	3.18	1.51	1.39	0.45	0.110	0.293
SoftPlatt	—	1.52	1.31	**0.35**	0.110	0.288
VS	—	1.35	1.26	0.39	**0.109**	0.289
Dirichlet-Soft	—	1.25	1.06	**0.35**	**0.109**	**0.271**
IR-Soft	—	**0.72**	**0.83**	0.45	0.115	0.340
ViT-B/16 — Voted-label baselines						
Uncalibrated	—	4.99	5.36	1.06	0.111	0.678
TS	2.04	4.48	4.40	0.93	0.106	0.346
Platt (PS)	—	4.54	4.38	0.93	0.105	0.363
Dirichlet-Hard	—	4.70	4.62	0.97	0.107	0.394
HB-Hard	—	6.55	6.44	5.84	0.388	7.37[†]
ViT-B/16 — Soft-label methods (ours)						
MCTS	3.07	0.85	0.57	0.39	0.102	0.278
SLTS	3.07	0.85	0.56	0.39	0.102	0.278
SoftPlatt	—	0.88	0.47	**0.24**	0.101	0.272
VS	—	0.92	0.50	0.32	0.101	0.274
Dirichlet-Soft	—	**0.72**	**0.41**	0.25	**0.100**	**0.255**
IR-Soft	—	0.91	0.61	0.43	0.105	0.321

[†] HB assigns $(1-c)/(K-1)$ to non-predicted classes; large NLL/Br for $K = 10$.

Table 12 provides the full numerical values underlying Figure 2, showing ECE broken down by annotation entropy quartile across all methods.

L Implementation Details

CIFAR-10H models. *ResNet-50*: ImageNet-1k weights (`ResNet50_Weights.IMAGENET1K_V1`); FC layer replaced with 2048×10 linear; AdamW, lr= 10^{-4}, weight decay 10^{-4}, batch 128, cosine annealing, 30 epochs, 224×224. *ViT-B/16*: ImageNet-21k weights; classification head replaced; AdamW, lr= 5×10^{-5}, weight decay 10^{-4}, batch 64, cosine annealing, 20 epochs, 224×224.

ChaosNLI models. *RoBERTa-Large*: `roberta-large-mnli` from HuggingFace, pre-fine-tuned on MultiNLI [Liu et al., 2019]; no additional fine-tuning. *DeBERTa-v3-base*: `cross-encoder/nli-deberta-v3-base` from HuggingFace [He et al., 2023]; no additional fine-tuning. Both models use the combined ChaosNLI SNLI+MNLI subset ($n = 3{,}113$), split 50/50 into calibration and test (stratified, seed 42).

ISIC 2019 models. *EfficientNet-B4*: ImageNet-1k weights; stratified 70/15/15 split; class-weighted cross-entropy (NV: $\approx 67\%$ of training images); AdamW, lr= 3×10^{-4}, weight decay 10^{-4}, 2-epoch linear warm-up + cosine decay, 20 epochs, 380×380. *ViT-S/16*: ImageNet-21k weights; same split and loss weighting; AdamW, lr= 10^{-4}, weight decay 10^{-4}, 2-epoch warm-up + cosine decay, 20 epochs, 224×224.

Table 11: **Full results on ChaosNLI** (combined SNLI+MNLI, 100 annotators per example). Metrics as in Table 10.

Method	T	ECE (%) ↓	aECE (%) ↓	cwECE (%) ↓	Br ↓	NLL ↓
RoBERTa-Large — Voted-label baselines						
Uncalibrated	—	27.79	27.77	18.61	0.650	1.384
TS	2.42	10.55	10.38	7.30	0.537	0.886
Platt (PS)	—	11.16	10.92	7.28	0.530	0.875
Dirichlet-Hard	—	11.55	11.40	7.64	0.535	0.889
HB-Hard	—	8.37	8.22	8.33	0.562	0.963
RoBERTa-Large — Soft-label methods (ours)						
MCTS	3.41	3.23	2.59	4.62	0.520	0.862
SLTS	3.41	3.22	2.59	4.62	0.520	0.862
SoftPlatt	—	2.66	2.32	2.43	**0.509**	**0.839**
VS	—	3.46	3.07	5.13	0.518	0.857
Dirichlet-Soft	—	**2.57**	**1.71**	**2.03**	0.510	**0.839**
IR-Soft	—	2.65	2.59	6.24	0.549	0.934
DeBERTa-v3-base — Voted-label baselines						
Uncalibrated	—	35.97	35.92	24.02	0.743	2.148
TS	3.81	11.63	11.57	8.48	0.549	0.900
Platt (PS)	—	12.43	12.36	8.43	0.539	0.896
Dirichlet-Hard	—	12.69	12.63	8.84	0.539	0.895
HB-Hard	—	10.75	10.26	9.71	0.574	0.989
DeBERTa-v3-base — Soft-label methods (ours)						
MCTS	5.27	3.49	3.38	6.56	0.529	0.876
SLTS	5.28	3.45	3.36	6.53	0.529	0.877
SoftPlatt	—	3.17	2.92	3.18	0.511	0.841
VS	—	3.91	3.35	6.46	0.525	0.869
Dirichlet-Soft	—	3.19	**2.71**	**2.95**	**0.509**	**0.836**
IR-Soft	—	**2.15**	3.96	7.20	0.554	0.942

Table 12: **ECE by annotation entropy quartile (CIFAR-10H).** All values in %. Q1 = lowest ambiguity; Q4 = highest.

	Method	Q1	Q2	Q3	Q4
Voted-label	Uncalibrated	1.9	4.2	8.7	21.3
	TS	1.4	4.8	11.2	18.1
	Platt (PS)	1.5	4.5	10.8	17.6
Soft (ours)	MCTS	1.5	2.9	4.1	7.3
	SLTS	1.3	2.4	3.8	**6.2**
	IR-Soft	**1.1**	**2.3**	**3.5**	6.7

DermaMNIST models. *ResNet-18*: ImageNet-1k weights, class-weighted cross-entropy, AdamW, lr$= 10^{-4}$, 30 epochs. *ViT-S/16*: ImageNet-21k weights, same training protocol, 224×224.

Calibration optimizers. TS, SLTS, MCTS: LBFGS, lr$= 0.1$, 500 iterations, tolerance 10^{-9} (gradient), 10^{-11} (loss). Platt/VS: Adam, lr$= 0.01/0.05$, weight decay 10^{-4}, 2000 steps.

ECE bins. $B = 15$ equal-width bins in $[0, 1]$; bins with fewer than 1 example are excluded.

MCTS. $S = 50$ samples per calibration example, fixed seed 42. Sensitivity to S is negligible for $S \geq 20$.

Compute. CIFAR-10H fine-tuning: ≈20 min on one NVIDIA A100. All calibration methods: < 60 s on CPU after logit caching.